



SE431 SOFTWARE QUALITY ASSURANCE

# MicroLend

Community Microfinance Loan Servicing and Delinquency Engine

MILESTONE 1 · WEEK 6

Quality Requirements, RTM & SQA Plan

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[github.com/tughral1/microlend](https://github.com/tughral1/microlend)

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**Companion document:** <docs/M1-SE423-Proposal-000-Pattern-Plan.md> — the construction-side proposal this quality plan applies to.

## 1 Quality Requirements (ISO/IEC 25010)

### 1.1 Which characteristics matter for this system, and which do not

ISO/IEC 25010:2011, the version used in this course, defines eight product quality characteristics. MicroLend's product risk is **financial, not safety-related**, and it is an internal back-office tool with a handful of users, which determines where quality effort is spent:

| Characteristic                | Relevance to MicroLend | Why                                                                                                                                                 |
|-------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Functional suitability</b> | <b>Critical</b>        | A schedule or allocation that is arithmetically wrong corrupts every downstream figure — balances, penalties, classification, provisioning, reports |
| <b>Reliability</b>            | <b>Critical</b>        | A failure part-way through posting must not leave a half-written ledger; the end-of-day engine must survive a task dying                            |
| <b>Security (integrity)</b>   | <b>High</b>            | A ledger that can be altered is not an audit record. Integrity, not confidentiality, is the sub-characteristic that matters                         |
| <b>Maintainability</b>        | <b>High</b>            | The system is assessed across three milestones by three developers who must each work in code the others wrote                                      |
| <b>Performance efficiency</b> | <b>Moderate</b>        | The end-of-day run must finish within an overnight window. Nothing else is time-critical                                                            |
| <b>Usability</b>              | <b>Low</b>             | A small internal web interface used by trained branch staff on a local machine. Learnability is not a project risk                                  |
| <b>Compatibility</b>          | <b>Low</b>             | Interoperability only at the bank-statement import (FR-20) and the CSV export (FR-32), both verified functionally; no co-existence requirement      |
| <b>Portability</b>            | <b>Low</b>             | Adaptability only between members' Windows laptops and the Linux CI runner, verified by every CI build; the system is never deployed                |

No characteristic is quietly omitted: compatibility and portability are rated **Low**, with the reason stated.

### 1.2 The quality requirements

Each requirement is tied to one characteristic, states a measurable target, and names one of the four verification methods — **inspection, analysis, demonstration, test**.

| ID           | Requirement                                                                                                       | Characteristic         | Sub-characteristic | Measurable target                                                      | Verification method and evidence                                                                                  |
|--------------|-------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>QR-01</b> | End-of-day processing of 10,000 active loans shall complete within 20 seconds on the demo machine using 4 threads | Performance efficiency | Time behaviour     | Median wall-clock over 5 consecutive runs ≤ 20 s; no single run > 25 s | <b>Test</b> — timed JUnit harness over a generated 10,000-loan dataset; timings attached to each milestone report |

|              |                                                                                                                        |                        |                               |                                                                                                                                                                                                                                                     |                                                                                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QR-02</b> | Schedule generation shall reproduce the hand-computed reference schedules exactly, at the stated monetary scale        | Functional suitability | Functional correctness        | Zero discrepancy against 30 hand-computed golden schedules (10 per product) at scale 2, <code>HALF_EVEN</code>                                                                                                                                      | <b>Test</b> — golden-file unit tests ( <code>GoldenScheduleTest</code> ). Allocation correctness is covered separately by the 100% decision-coverage policy on the money path (SQAP §4.4) |
| <b>QR-03</b> | Any failure during repayment posting shall leave the ledger balanced, with no partial entry                            | Reliability            | Fault tolerance               | For 100% of at least 10 injected failure points: $\Sigma \text{ debits} = \Sigma \text{ credits}$ , and zero orphan entries                                                                                                                         | <b>Test</b> — fault injection through a repository stub that throws at each write point; also <b>demonstrated</b> live at M3 as one of the two required failure scenarios                 |
| <b>QR-04</b> | Core business-logic methods shall remain within the stated complexity and size limits                                  | Maintainability        | Analysability / Modifiability | Mean cyclomatic complexity $\leq 5$ per method across the five module packages, computed from JaCoCo's per-method complexity counter; no method above 15 and no source file above 400 lines (the Checkstyle build gate, §6.5 of the SE423 document) | <b>Analysis</b> — static-analysis metric report produced in CI and attached to each milestone                                                                                             |
| <b>QR-05</b> | A posted ledger entry shall not be alterable by any component; corrections shall occur only as compensating entries    | Security               | Integrity                     | Zero mutating members on <code>LedgerEntry</code> ; zero <code>UPDATE</code> or <code>DELETE</code> statements against <code>ledger_entry</code> in the codebase; 100% of corrections implemented as compensating entries                           | <b>Inspection</b> — an explicit item on the Fagan inspection checklist, supported by a static-analysis rule and a unit test asserting reversal-only correction                            |
| <b>QR-06</b> | The end-of-day engine shall produce identical results whether or not repayments are posted concurrently during the run | Reliability            | Maturity                      | Over 50 repeated runs of the concurrency suite: zero lost updates, zero double accruals, identical final balances to the single-threaded reference run                                                                                              | <b>Test</b> — deterministic latch-driven concurrency tests, repeated 50 times by a scheduled CI job added at M2 with the engine itself                                                    |

### 1.3 Constraints (not quality requirements)

The standard separates requirements into functional, non-functional (quality) and **constraints**. The following are constraints on how the system is built. They are recorded in SQAP §4.4 rather than presented as quality requirements:

| ID          | Constraint                                                                                                        | Rationale                                                                                                                                                                                          |
|-------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C-01</b> | Monetary values shall never be represented as <code>double</code> or <code>float</code>                           | <code>0.1 + 0.2 <math>\neq</math> 0.3</code> in IEEE 754; a fraction of a rupee per instalment accumulates into a ledger that does not balance. Enforced by a Checkstyle rule that fails the build |
| <b>C-02</b> | <code>BigDecimal</code> values shall be compared with <code>compareTo() == 0</code> , never <code>equals()</code> | <code>equals()</code> also compares scale, so <code>2.50</code> is not equal to <code>2.5</code>                                                                                                   |
| <b>C-03</b> | No cyclic dependencies between module packages                                                                    | Cycles destroy the ability to test or replace a module in isolation. Enforced from M2 by an ArchUnit test in CI                                                                                    |

**C-04** Monetary columns shall be stored as integer minor units      Same reasoning as C-01, applied at the storage layer

## 1.4 The quality trade-off

**QR-01 (performance) is in tension with QR-03, QR-05 and QR-06 (fault tolerance, integrity, maturity).** Per-account locking, immutable value objects, append-only correction and an idempotency key all cost time behaviour: each adds work on the path that QR-01 measures.

The conflict is resolved by **capping the cost rather than removing either attribute** — locks are per account rather than global, so unrelated accounts are computed in parallel, and each lock is held only for that account's own short transaction. The measured cost is recorded in the GQM data at each milestone rather than asserted to be small.

# 2 Requirements Traceability Matrix — Version 1

## 2.1 Purpose and how it will be maintained

The RTM supports traceability in **both directions**. Forward — from requirement to code to test — answers “*did we build and verify everything that was asked for?*” and exposes a **test gap**: a requirement implemented but never verified. Backward — from code to requirement — answers “*why does this code exist?*” and exposes **rogue code**: a module traceable to no approved requirement.

Those are the two defects this matrix is audited for at every milestone. At M1 the design and class columns are planned rather than built, so they are marked planned; from M2 they name real artifacts and real test results.

The matrix is maintained at [qa/rtm.md](#) in the repository and versioned with the code, so the state of traceability at any baseline can be reproduced from the tag.

## 2.2 Functional requirements

Each requirement traces to a rule in the Business Rules Register ( [docs/business-rules.md](#) ). The three requirements marked “—” (FR-20, FR-31, FR-32) have no business rule because they are interface or operational requirements, not lending rules.

| ID           | Requirement                                                                                                                                | Business rule | Module                      | Design element (planned)            | Planned test — level and technique                                           |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------|-------------------------------------|------------------------------------------------------------------------------|
| <b>FR-01</b> | An applicant shall be at least 18 years old at disbursement and no more than 65 at maturity                                                | BR-01, BR-02  | <a href="#">origination</a> | <a href="#">AgeRule</a>             | Unit — <b>BVA</b> : 17 / 18 at disbursement; 64 / 65 / 66 at maturity        |
| <b>FR-02</b> | An applicant shall hold no more than 2 active loans, counting the one applied for (a loan in arrears is active; a written-off loan is not) | BR-03         | <a href="#">origination</a> | <a href="#">ActiveLoanLimitRule</a> | Unit — <b>EP + BVA</b> : 0, 1, 2, 3 existing active loans (2 and 3 rejected) |
| <b>FR-03</b> | Total monthly instalment burden, including the new loan, shall not exceed 40% of declared monthly income                                   | BR-04         | <a href="#">origination</a> | <a href="#">BurdenRatioRule</a>     | Unit — <b>BVA</b> : 39.99% / 40.00% / 40.01%                                 |
| <b>FR-04</b> | An applicant with a prior write-off shall be rejected                                                                                      | BR-05         | <a href="#">origination</a> | <a href="#">WriteOffHistoryRule</a> | Unit — <b>EP</b> : no history / written-off / settled late                   |

|              |                                                                                                                                                         |                 |             |                                                                                       |                                                                                                                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>FR-05</b> | A rejected application shall state a typed reason naming the first rule that failed                                                                     | BR-08           | origination | EligibilityEvaluator , Decision , RejectionReason                                     | Unit — <b>EP</b> : one case per rejection reason                                                                     |
| <b>FR-06</b> | Risk grade A/B/C shall be assigned from the burden ratio and repayment history, and shall adjust the rate by a fixed spread and cap the approved amount | BR-09,<br>BR-10 | origination | RiskGrade , Decision                                                                  | Unit — <b>EP + BVA</b> : one case per grade; burden 25.00 / 25.01 / 35.00 / 35.01%; worst past DPD 30 / 31 / 60 / 61 |
| <b>FR-07</b> | The requested amount and tenure shall lie within the product's bands                                                                                    | BR-06,<br>BR-07 | origination | ProductBandRule                                                                       | Unit — <b>BVA</b> : min – 1 / min / max / max + 1, for amount and tenure                                             |
| <b>FR-08</b> | The system shall generate schedules for flat-rate, reducing-balance and bullet products                                                                 | BR-11,<br>BR-17 | products    | LoanProduct and its three subclasses;<br>InterestCalculator and its three calculators | Unit — <b>Golden file</b> : 10 hand-computed schedules per product                                                   |
| <b>FR-09</b> | Instalment dates shall follow the disbursement day-of-month, clamped for short months                                                                   | BR-12           | products    | DueDateCalculator                                                                     | Unit — <b>BVA</b> : disbursement on the 28th, 29th, 30th, 31st; February in leap and non-leap years                  |
| <b>FR-10</b> | A grace period of up to 3 months shall defer the first instalment while interest still accrues                                                          | BR-13           | products    | GracePeriodPolicy                                                                     | Unit — <b>EP + BVA</b> : grace 0 / 1 / 3 / 4 (4 rejected)                                                            |
| <b>FR-11</b> | The instalments of a schedule shall sum exactly to principal plus total interest                                                                        | BR-14,<br>BR-15 | products    | ScheduleAssembler , Schedule , Instalment                                             | Unit — <b>Property-based invariant</b> over valid amount/tenure/rate combinations                                    |
| <b>FR-12</b> | A schedule shall contain exactly <code>tenureMonths</code> instalments                                                                                  | BR-16           | products    | ScheduleAssembler                                                                     | Unit — <b>EP</b> : tenure 1 / 6 / 36, with and without grace                                                         |
| <b>FR-13</b> | Repayments shall be applied in the order penalty → fees → interest → principal, oldest instalment first                                                 | BR-18,<br>BR-19 | ledger      | RepaymentAllocator , method <code>allocate()</code> ; Allocation                      | Unit — <b>Basis path</b> (CC ≈ 11) + EP on payment size                                                              |
| <b>FR-14</b> | A partial payment shall be applied as far as it reaches, leaving the remainder outstanding                                                              | BR-20           | ledger      | RepaymentAllocator                                                                    | Unit — <b>BVA</b> : 0 / 0.01 / due – 0.01 / exactly due                                                              |
| <b>FR-15</b> | An overpayment shall become an advance credit unless it clears the loan, in which case early settlement applies                                         | BR-21,<br>BR-22 | ledger      | SettlementPolicy                                                                      | Unit — <b>BVA</b> : due + 0.01 / exact payoff / payoff + 0.01, per product                                           |
| <b>FR-16</b> | Every posting shall write balanced debit and credit lines                                                                                               | BR-23           | ledger      | CoreRepaymentPoster                                                                   | Unit — <b>Invariant</b> : $\sum \text{debits} = \sum \text{credits}$ after every posting                             |

|              |                                                                                                    |                 |                  |                                                        |                                                                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------|-----------------|------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-17</b> | A posted ledger entry shall be immutable; corrections occur only as compensating entries           | BR-24           | ledger           | LedgerEntry ,<br>CompensatingEntryService              | Unit + inspection — checklist item + test asserting no mutator exists                                                                               |
| <b>FR-18</b> | A payment whose (bank, reference) pair is already in the ledger shall be rejected as a duplicate   | BR-25           | ledger           | IdempotencyGuardPoster                                 | Unit — <b>EP</b> : new reference / same reference twice                                                                                             |
| <b>FR-19</b> | A payment against a closed or written-off loan shall be rejected with a typed exception            | BR-26           | ledger           | LoanAccount ,<br>LoanClosedException                   | Unit — <b>EP</b> : active / closed / written-off                                                                                                    |
| <b>FR-20</b> | Repayments shall be importable from bank statement files in each supported bank's format           | —               | ledger.importing | HblStatementAdapter ,<br>McbStatementAdapter           | Unit + integration — <b>EP</b> : valid row / malformed date / minor-unit amount / duplicate reference                                               |
| <b>FR-21</b> | Penalty shall accrue at 0.05% per day on overdue principal, capped at 10% of the overdue amount    | BR-27,<br>BR-28 | delinquency      | PenaltyAccruer                                         | Unit — <b>BVA</b> : just below / at / above the cap (unit level: under BR-32 the cap cannot be reached before write-off); <b>EP</b> on days overdue |
| <b>FR-22</b> | Days past due shall be counted from the day after the instalment due date                          | BR-29           | delinquency      | DaysPastDueCalculator                                  | Unit — <b>BVA</b> : due date / due + 1                                                                                                              |
| <b>FR-23</b> | Accounts shall be bucketed 0 / 1-30 / 31-60 / 61-90 / 91+ days past due and classified accordingly | BR-30           | delinquency      | DelinquencyClassifier ,<br>method classify()           | Unit — <b>Basis path</b> (CC ≈ 6) + <b>BVA</b> : 0/1/30/31/60/61/90/91                                                                              |
| <b>FR-24</b> | A provision percentage shall be applied per classification                                         | BR-31           | delinquency      | ProvisioningPolicy                                     | Unit — <b>EP</b> : one case per classification                                                                                                      |
| <b>FR-25</b> | A loan more than 180 days past due shall be written off automatically                              | BR-32           | delinquency      | WriteOffPolicy                                         | Unit — <b>BVA</b> : 179 / 180 / 181 days                                                                                                            |
| <b>FR-26</b> | End-of-day processing shall be idempotent per business date                                        | BR-33           | delinquency      | EndOfDayEngine ,<br>delinquency_snapshot<br>unique key | Unit — <b>Invariant</b> : the same date run twice changes nothing                                                                                   |
| <b>FR-27</b> | A classification change shall raise an alert event                                                 | BR-34           | delinquency      | ClassificationChangeEvent                              | Unit — <b>EP</b> : downgrade / upgrade / no change                                                                                                  |

|              |                                                                                                                      |                  |           |                                                        |                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------|------------------|-----------|--------------------------------------------------------|--------------------------------------------------------------------------|
| <b>FR-28</b> | The system shall produce a customer statement for a date range, with a running balance                               | BR-35            | reporting | StatementRenderer                                      | Unit — <b>EP</b> : empty range / single entry / full history             |
| <b>FR-29</b> | The system shall report portfolio at risk over 30 days, provision coverage and collection efficiency                 | BR-36 –<br>BR-38 | reporting | PortfolioReporter                                      | Unit — <b>Golden file</b> : a fixed portfolio with hand-computed metrics |
| <b>FR-30</b> | Reports shall be reproducible as at any past business date                                                           | BR-39            | reporting | PortfolioReporter                                      | Unit — <b>EP</b> : current date / past date / date before the first loan |
| <b>FR-31</b> | Every teller posting shall record an audit journal line naming the operator; a bank import records one line per file | —                | ledger    | AuditLoggingPoster (teller); the import job (per file) | Unit — <b>EP</b> : teller posting / imported file                        |
| <b>FR-32</b> | Portfolio reports shall be exportable as CSV                                                                         | —                | reporting | ReportExporter ,<br>CsvReportExporter                  | Unit — <b>EP</b> : empty portfolio / one loan / full portfolio           |

## 2.3 Quality requirements traced

| ID    | Verification method  | Primary artifact (planned)                                                                  | Evidence produced at |
|-------|----------------------|---------------------------------------------------------------------------------------------|----------------------|
| QR-01 | Test                 | EndOfDayPerformanceTest                                                                     | M2, M3               |
| QR-02 | Test                 | GoldenScheduleTest                                                                          | M2, M3               |
| QR-03 | Test + demonstration | PostingFaultInjectionTest                                                                   | M2, demonstrated M3  |
| QR-04 | Analysis             | Static-analysis metric report in CI                                                         | M2, M3               |
| QR-05 | Inspection           | Fagan inspection checklist item + LedgerImmutabilityTest                                    | M2                   |
| QR-06 | Test                 | EndOfDayConcurrencyTest over EndOfDayEngine and LockRegistry , 50 runs per scheduled CI job | M2, M3               |

## 2.4 Coverage of the matrix at M1

| Measure                                            | Status at M1                                                                                                                           |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Functional requirements defined                    | 32                                                                                                                                     |
| Quality requirements defined                       | 6                                                                                                                                      |
| Constraints recorded                               | 4                                                                                                                                      |
| Requirements with a planned verification approach  | 38 of 38 (100%)                                                                                                                        |
| Requirements with executed tests                   | 0 — M1 is a planning milestone                                                                                                         |
| Planned design elements traceable to a requirement | Every planned business-logic class in the SE423 class views traces to an FR or QR; the rest are shared kernel and infrastructure types |
| Code modules traceable to a requirement            | n/a at M1; audited from M2                                                                                                             |

**Audit rule for M2 and M3:** every requirement must have at least one test (no test gaps), and every production class must trace to at least one requirement (no rogue code). Both directions are checked, because neither direction can find the other's defect.

### 3 SQA Plan — Scope and Outline (IEEE 730)

The outline below follows the SQAP outline in Annex C of IEEE 730-2014. Every mandatory section is listed with a one-line plan specific to MicroLend. M1 establishes the scope and the structure; the plan is developed through M2 and completed as the final SQAP at M3 (Week 13), checked against the final build.

| §     | IEEE 730 section                     | Plan for MicroLend                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Purpose and scope                    | This SQAP governs the MicroLend loan-servicing system across M1–M3, covering the five modules and the shared repository, tailored to a three-person Very Small Entity                                                                                                                         |
| 2     | Definitions and acronyms             | Define the terms actually used — DPD, PAR30, allocation waterfall, EOD, compensating entry, cyclomatic complexity, EP/BVA — plus error, defect and failure, so the document is self-contained                                                                                                 |
| 3     | Reference documents                  | The joint-project guidelines (updated 7 October 2026: three milestones), ISO/IEC 25010:2011, ISO/IEC/IEEE 29148, IEEE 730-2014, IEEE 1012 (V&V plan at M2), IEEE 1028, ISO/IEC 29110, the Business Rules Register and the RTM                                                                 |
| 4     | SQA plan overview                    | Organisation, product risk, tools, standards and effort for MicroLend — 4.1 to 4.5 below                                                                                                                                                                                                      |
| 4.1   | Organisation and independence        | Name the three members, their module ownership and the rotating SQA lead. State that organisational independence is impossible in a three-person team, and that it is substituted by <b>non-owner review</b> (no one approves their own pull request) and the <b>external inspector at M2</b> |
| 4.2   | Software product risk                | Classify MicroLend's product risk as <b>financial, not safety-related</b> ; identify the money path ( <code>RepaymentAllocator</code> , <code>DelinquencyClassifier</code> , the ledger) as the high-risk area; set SQA depth from that, since SQA effort is proportionate to product risk    |
| 4.3   | Tools                                | Java 21 and JUnit 5, SQLite, Gradle, GitHub and GitHub Actions, JaCoCo for coverage, Checkstyle as the coding-standard gate, ArchUnit for the no-cycles constraint, and a second static analyser for the M2 report — each named with the quality question it answers                          |
| 4.4   | Standards, practices and conventions | Constraints C-01 to C-04; the commit convention <code>test(red): / feat(green): / refactor: ;</code> the documentation standard; and the <b>100% decision coverage on the money path</b> policy                                                                                               |
| 4.5   | Effort, resources and schedule       | About 39 student-days of construction in total, plus about 8 student-days of SQA per milestone (24 across three), so about 63 student-days — roughly three person-months — mapped onto the Week 6 / 10 / 13 milestone dates with named owners                                                 |
| 5     | Activities, outcomes and tasks       | Product assurance (5.1) and process assurance (5.2), each tied to a milestone deliverable                                                                                                                                                                                                     |
| 5.1   | Product assurance                    | Evaluate the plans, the product and its support documentation, and measure the product — 5.1.1 to 5.1.5                                                                                                                                                                                       |
| 5.1.1 | Evaluate plans for conformance       | At each milestone, check that the M1 proposal, the test plan and this SQAP still describe what is actually being built; record deviations                                                                                                                                                     |

|       |                                                     |                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.1.2 | Evaluate product for conformance                    | The Fagan inspection of the two money-path classes, static-analysis triage, and black-box and white-box test execution against the RTM requirement IDs                                                                                                                                                                                                                      |
| 5.1.3 | Evaluate product for acceptability                  | Define the acceptance gate applied to the final integrated system at M3: all functional requirements traced to passing tests, $\geq 60\%$ coverage of core logic with 100% decision coverage on the money path, zero open critical defects                                                                                                                                  |
| 5.1.4 | Evaluate product life-cycle support for conformance | Check at each baseline that the README, setup instructions, schema DDL and design documents match the code                                                                                                                                                                                                                                                                  |
| 5.1.5 | Measure products                                    | Defect density per module, branch coverage, cyclomatic complexity per method, open and closed defect counts — the metrics the GQM questions consume                                                                                                                                                                                                                         |
| 5.2   | Process assurance                                   | Evaluate the life-cycle processes, environments and subcontractor processes, measure the processes, and assess staff skills — 5.2.1 to 5.2.5                                                                                                                                                                                                                                |
| 5.2.1 | Evaluate life-cycle processes for conformance       | Verify that the red-green-refactor sequence, pull-request review and merge gates were genuinely followed, using commit and pull-request history as the evidence                                                                                                                                                                                                             |
| 5.2.2 | Evaluate environments for conformance               | Confirm all three members build and test on the same toolchain versions, and that CI is the single source of truth for a green build                                                                                                                                                                                                                                        |
| 5.2.3 | Evaluate subcontractor processes for conformance    | <b>Declared not applicable, with justification:</b> there are no subcontractors. Third-party components are limited to open-source libraries (the SQLite JDBC driver, Apache Commons CSV, and test libraries), pinned to exact versions in <code>build.gradle</code> as each is added, and checked by CI on every build. The standard requires the declaration, not silence |
| 5.2.4 | Measure processes                                   | Inspection rate in LOC per hour, defect-removal effectiveness by phase, and rework as a percentage of total effort                                                                                                                                                                                                                                                          |
| 5.2.5 | Assess staff skill and knowledge                    | Record that concurrency, white-box test design, static analysis and the Fagan inspection process are <b>not yet taught</b> at M1, and name the self-study and pairing that closes each gap before the technique is used                                                                                                                                                     |
| 6     | Additional considerations                           | Contract review, quality measurement, waivers, task repetition, SQA risk, communication and non-conformance — 6.1 to 6.7                                                                                                                                                                                                                                                    |
| 6.1   | Contract review                                     | Treat the joint-project guidelines as the agreement: a checklist mapping every Section 3.1 and 3.2 minimum to where it is met, re-checked at each milestone                                                                                                                                                                                                                 |
| 6.2   | Quality measurement                                 | The GQM programme — one goal, three questions, and the metrics listed in §5.1.5 and §5.2.4                                                                                                                                                                                                                                                                                  |
| 6.3   | Waivers and deviations                              | Any departure from this plan is logged with reason, approver and date — for example, dropping a planned pattern is recorded, not silently abandoned                                                                                                                                                                                                                         |
| 6.4   | Task repetition                                     | A re-inspection is triggered if an inspection finds more than five major defects in the roughly 250-line inspection package; the full regression suite re-runs on every push                                                                                                                                                                                                |
| 6.5   | Risk to performing SQA                              | Risks to the quality work itself: exam weeks compressing the inspection, the external inspector falling through, one member carrying the SQA load                                                                                                                                                                                                                           |
| 6.6   | Communications strategy                             | A weekly 15-minute session where each member explains their module's invariants to the other two; defect log and corrective actions in the repository; milestone status summarised in the M2 and M3 documents                                                                                                                                                               |
| 6.7   | Non-conformance process                             | How a defect or process non-conformance is raised, triaged by severity, assigned, corrected and verified — the corrective-action (CAPA) loop opened at M2 with the inspection and static-analysis findings, and closed — or kept open with a stated reason — in the M3 final audit                                                                                          |
| 7     | SQA records                                         | What is recorded and where it is kept — 7.1 and 7.2                                                                                                                                                                                                                                                                                                                         |

|     |                                      |                                                                                                                                                                                                          |
|-----|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1 | SQA records: collect, file, maintain | All quality records live in <code>/qa</code> in the repository: inspection log, defect log with corrective-action status, static-analysis reports, coverage reports, GQM data and risk-register versions |
| 7.2 | Availability of records              | Records are versioned with the code in the hosted repository, so any record can be produced on request at the demo                                                                                       |

### 3.1 Inspection planning note (for M2)

The inspection runs in six stages: **planning** → **overview (kickoff)** → **preparation (individual checking)** → **inspection (logging) meeting** → **rework (edit)** → **follow-up and exit**. The planning guideline is **100–200 lines of code per hour** for source code.

That rate is the number to plan with. Inspecting `RepaymentAllocator` plus `DelinquencyClassifier` — roughly 250 lines — means a logging meeting of about 1¼ to 2½ hours — split into two sessions of at most two hours if the slower rate applies — which is schedulable with an external inspector and which also provides the denominator for the defect-detection-rate metric in the GQM programme. No member moderates their own code: Muhammad Ibrahim moderates `RepaymentAllocator` (Tughrul Hussain's module), Hassan Khalid moderates `DelinquencyClassifier` (Muhammad Ibrahim's), and the external inspector reviews both.

## 4 Cost of Quality and Quality-Culture Statement

### 4.1 The model

Total cost of quality is the sum of the **cost of conformance** (what is spent to prevent and detect defects) and the **cost of non-conformance** (what defects cost once they exist).

### 4.2 Expected costs for this project, in student-hours

The currency here is time, because that is the resource the team actually spends. Figures are estimates made at M1 and compared against actuals at M3 — the comparison is itself a GQM metric.

| Category          | Activity for MicroLend                                                  | Estimated effort (student-hours) |
|-------------------|-------------------------------------------------------------------------|----------------------------------|
| <b>Prevention</b> | Writing the Business Rules Register and freezing scope                  | 6                                |
|                   | Coding standard and Checkstyle configuration (constraints C-01 to C-04) | 4                                |
|                   | Hand-computing 30 golden schedules in a spreadsheet before coding       | 8                                |
|                   | Design review of the class model before implementation                  | 6                                |
|                   | Setting up CI, branch protection and quality gates                      | 5                                |
|                   | <b>Prevention subtotal</b>                                              | <b>29</b>                        |
| <b>Appraisal</b>  | Writing and maintaining the test suite (test-first)                     | 60                               |
|                   | Pull-request review by a non-owner                                      | 18                               |
|                   | Fagan inspection of the money path, including preparation               | 12                               |
|                   | Static-analysis runs and triage                                         | 8                                |
|                   | <b>Appraisal subtotal</b>                                               | <b>98</b>                        |

|                                       |                                                                                          |                                           |
|---------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Internal failure</b>               | Rework of defects found by tests, review or inspection before submission                 | 25 (expected)                             |
| <b>External failure</b>               | Rework of defects found by the grader at a milestone demo, or by the following milestone | 10 (target: as close to zero as possible) |
| <b>Non-conformance subtotal</b>       |                                                                                          | <b>35</b>                                 |
| <b>Total expected cost of quality</b> |                                                                                          | <b>162 student-hours</b>                  |

### 4.3 What the team expects these numbers to show

Three things follow from this table, and each is a prediction the team is prepared to be held to at M3:

- 1. Appraisal dominates, and that is correct for this project.** The money path has no safety net other than tests: a wrong instalment does not crash, it quietly produces a wrong number. Detection effort is where the risk is.
- 2. Internal failure cost rising is a good sign, not a bad one.** If the register shows more hours of rework triggered by our own tests and inspections at M2 than at M1, it means defects are being caught inside the team rather than at a demo. The number to minimise is **external** failure, not internal. A team reporting zero internal failure cost is reporting that it is not looking.
- 3. The 1:10:100 principle is why the 8 hours on golden schedules are spent first.** A rounding rule corrected while writing the spreadsheet costs minutes. The same defect found after schedules, ledger entries, delinquency buckets and portfolio reports have all been built on it requires reworking every one of them, under deadline. Prevention is bought at the cheapest point on the curve.

### 4.4 Quality-culture statement

The team commits to four behaviours, each with a mechanism that makes it checkable rather than aspirational:

| Commitment                                                                                                                        | The mechanism that enforces it                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quality is not a phase at the end.</b> Quality activities run in the same milestone cycle as the construction work they assess | SQA deliverables are owned per milestone and rotate; the M2 inspection targets M2 code, not a frozen artifact                                        |
| <b>No one marks their own homework.</b> Every change is reviewed by someone who does not own that module                          | Branch protection requires one approving review from a non-author; <code>CODEOWNERS</code> routes the request                                        |
| <b>Defects are data, not blame.</b> A defect found in inspection is a success for the process                                     | The defect log records type, severity and phase found — never who wrote it. The metric is defect-removal effectiveness, not individual defect counts |
| <b>If it is not evidenced, it did not happen.</b> Every claim in this plan maps to an artifact in the repository                  | All quality records live in <code>/qa</code> , versioned with the code and reproducible from a milestone tag                                         |

## 5 Standards Awareness

**ISO/IEC/IEEE 12207** tells us *what processes must exist*. Of its processes we genuinely perform Configuration Management (branch protection, the `m1` – `m3` baseline tags) and Quality Assurance, which the 2017 edition groups as technical management processes, and Verification and Validation, two separate technical processes; our pull-request reviews and the Fagan inspection correspond to the 2008 edition's Software Review process. **IEEE 730-2014** tells us *how* the SQA function executes inside that lifecycle: its activities in three groups — SQA process implementation, product assurance, process assurance — its principle that the depth of SQA is set by **product risk**, and its rule that any activity not performed must be declared *not applicable with a justification* rather than quietly dropped. That rule is why our SQAP §5.2.3 explicitly marks subcontractor evaluation as not applicable instead of omitting it.

Full 12207 and CMMI adoption would be wrong at our scale. Three students on one project is a **Very Small Entity** under ISO/IEC 29110's 25-person definition, and at roughly three person-months of effort (\$4.5 of the SQAP outline) we fall within the **Entry** profile's six-person-month ceiling, though we follow the **Basic** profile's structure as taught. Its two processes map onto our milestone loop: Project Management covers planning, the risk register and milestone reviews, and Software Implementation covers the construction work each milestone delivers. Its Deployment Packages supply ready-made templates and traceability tables so our time goes into the work rather than into inventing document formats.

Against **CMMI-DEV v1.3** we claim **Maturity Level 2 behaviours only**, and we name them: Requirements Management through the RTM, Configuration Management through protected branches and baselines, Process and Product Quality Assurance through the SQAP §5.2.1 process-conformance checks and the M3 audit, and Measurement and Analysis through the GQM programme. We explicitly do **not** claim Level 3, because Level 3 requires an organisational set of standard processes, which a single student project does not have (the Fagan inspection is a peer review, which CMMI places in Verification at Level 3), and the statistical process control of Levels 4 and 5 is meaningless with three people and one data point per milestone.

MicroLend is not safety-critical, so no domain safety standard applies. We borrow one idea from them — criticality dictates technique — as the SQAP policy of **100% decision coverage on the money path** ( `RepaymentAllocator` , `DelinquencyClassifier` ), alongside the ban on `double` for money.

**Not claimed:** CMMI Level 3 or above; conformance to any safety-critical standard; organisational independence of the QA function; and certification of any kind. Every standard named above is used as a source of practice, not as a badge.

## 6 Initial Risk Register

The full register is maintained as a living document at `qa/risk-register.md` in the repository, updated at every milestone and whenever a risk materialises or closes. It is reproduced here as at M1.

### 6.1 Scoring

Risk exposure is **computed, not asserted**:  $E = P \times C$ , where P is probability (1 = rare ... 5 = almost certain) and C is consequence, i.e. impact (1 = negligible ... 5 = severe).

| Exposure | Priority | Response                                      |
|----------|----------|-----------------------------------------------|
| 17–25    | Critical | Act now; mitigation owned and tracked weekly  |
| 10–16    | High     | Mitigation planned and started this milestone |
| 5–9      | Medium   | Monitor; mitigate if exposure rises           |
| 1–4      | Low      | Accept; review each milestone                 |

### 6.2 The register at M1

P = probability, C = consequence, E = exposure. Residual exposure is recorded from M2, once each mitigation has produced evidence.

| ID | Risk (condition → consequence)                                                                                                                              | Type    | $P \times C = E$  | Priority | Mitigation                                                                                                                                                                                                                     | Owner            | Status |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------|
| R1 | Money or date arithmetic in schedule generation or the allocation waterfall is subtly wrong → every downstream figure is wrong and the defect is found late | Product | $4 \times 5 = 20$ | Critical | Hand-compute 30 golden schedules before coding and commit them as the first tests; <code>BigDecimal</code> only, enforced by constraint C-01; property test that $\Sigma$ instalments = principal + total interest             | Hassan Khalid    | Open   |
| R2 | M1 work starts late → the compressed schedule pushes work into the M2 window, and M2 (Week 10) carries 35%                                                  | Project | $5 \times 3 = 15$ | High     | Freeze scope at five modules with no new features; draft M1 documents from the existing analysis; set up repository and tooling in parallel with document writing                                                              | Tughrul Hussain  | Open   |
| R3 | The external inspector required at M2 is not arranged in time → the inspection criterion (20% of SE431 M2) cannot be fully evidenced                        | Process | $3 \times 4 = 12$ | High     | Request the instructor's team pairing immediately after M1; agree a date in Week 8; prepare the inspection package in advance                                                                                                  | Muhammad Ibrahim | Open   |
| R4 | Commits are squashed or tests are written after the code → the red-green-refactor trail cannot be reconstructed, losing TDD marks at M2 (10%) and M3 (15%)  | Process | $3 \times 4 = 12$ | High     | Squash-merge disabled in repository settings from day one; commit convention enforced; merges use <code>--no-ff</code> ; the pull-request template asks for the red commit hash                                                | Tughrul Hussain  | Open   |
| R5 | Concurrency defects in the end-of-day engine appear only under load → late, hard-to-reproduce failures near the M3 concurrency-correctness check            | Product | $3 \times 4 = 12$ | High     | Immutable value objects; per-account locks, one held at a time; a unique <code>(loan_id, business_date)</code> idempotency key in the same transaction as accrual; deterministic latch-driven tests with no <code>sleep</code> | Muhammad Ibrahim | Open   |
| R6 | Individual contribution is not attributable → individual evidence suffers, which is 20% of each member's milestone mark                                     | Project | $3 \times 3 = 9$  | Medium   | Module ownership equals directory ownership; every change by pull request reviewed by a non-owner; SE431 deliverable ownership rotates each milestone; AI Usage Log maintained continuously                                    | Tughrul Hussain  | Open   |
| R7 | Several M2-graded techniques are not taught until after they must be designed for → the team designs ahead of the lectures and gets it wrong                | Project | $4 \times 4 = 16$ | High     | Design the structures now, defer detailed technique work until taught; start the evidence streams that cannot be reconstructed later at M1; re-check each design against the lecture when delivered                            | Muhammad Ibrahim | Open   |
| R8 | AI-assisted work is submitted that a member cannot explain in the viva → reduced or zero individual marks, and a possible integrity concern                 | Project | $3 \times 5 = 15$ | High     | AI Usage Log updated continuously; the module owner reviews and rewrites generated code in their own module; dry-run Q&A inside the team before each demo                                                                      | Hassan Khalid    | Open   |

### 6.3 Review schedule

- **M1:** register initiated with eight risks, scored and owned.
- **M2 (Week 10):** every risk re-scored using evidence from the architecture, inspection, static analysis, testing and SCM; risks arising from implementation added; residual exposure recorded.
- **M3 (Week 13):** continuous updating from M1 to M3 demonstrated; mitigation effectiveness evaluated against actual project evidence; residual risk recorded for every risk.

## 7 Summary of M1 Quality Position

| Deliverable          | Status at M1                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality requirements | 6 defined, each with a measurable target and a named verification method; all eight characteristics rated, two of them Low with justification                         |
| Constraints          | 4 recorded, C-01 enforced by the build                                                                                                                                |
| RTM v1               | 32 functional and 6 quality requirements, each traced to a business rule where one exists and given a planned verification approach and a named test design technique |
| SQAP                 | Scope set and all mandatory IEEE 730 sections outlined with project-specific plans; one section declared not applicable with justification                            |
| Cost of Quality      | Estimated at 162 student-hours with the prevention/appraisal/failure split and three stated predictions                                                               |
| Standards awareness  | 12207, IEEE 730, 29110 and CMMI-DEV related to this project's scale, with explicit non-claims                                                                         |
| Risk register        | 8 risks, computed exposure, owners assigned, living document in the repository                                                                                        |

**What M1 deliberately does not yet contain**, because the techniques are taught later in the semester and M1 does not require them: executed test results, the inspection defect log, the static-analysis report, coverage figures and GQM data. Each has a planned home in this document and an owner. Of the four evidence streams that cannot be reconstructed afterwards, three are already running at M1 — the TDD commit trail, the SCM baselines and the CI static-analysis reports (kept for 90 days) — and GQM data collection starts with the first feature branch at M2.

# Annex A — AI Usage Log

Required by the joint-project guidelines §9. Milestones 2 and 3 carry their own logs.

Every entry used Claude through Claude Code. Each entry is dated the day the work was done, which the repository's commit history corroborates.

| # | Date       | Tool          | Purpose                                              | Prompt / activity summary                                                                                                      | Material adopted                                             | Modifications made                                                             | Verification                                                              |
|---|------------|---------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1 | 2026-10-04 | Claude Opus 5 | Compare candidate project ideas against both rubrics | Asked for candidate project ideas scored against the §3.1 and §3.2 minimum requirements                                        | Comparative analysis; MicroLend chosen over two alternatives | Scope cut to a locally-run application with no deployment or paid services     | Each rubric minimum traced to where it is met; team accepted the choice   |
| 2 | 2026-10-04 | Claude Opus 5 | Draft the SE423 document                             | Asked for a class model, SOLID analysis and pattern justifications from the module list and business rules                     | Class model, SOLID analysis, pattern justifications          | Diagram split into views for legibility; CLI replaced by a local web interface | Found and corrected a Simple Factory presented as a Factory Method        |
| 3 | 2026-10-04 | Claude Opus 5 | Draft the SE431 document                             | Asked for ISO 25010 quality requirements, an RTM, an IEEE 730 outline, a CoQ estimate and a risk register for the SE423 design | Quality requirements, RTM, SQAP outline, CoQ, risk register  | Four items reclassified from quality requirements to constraints               | Targets and effort estimates checked as realistic for a three-person team |
| 4 | 2026-10-04 | Claude Opus 5 | Repository setup and coding standard                 | Asked for a Checkstyle configuration enforcing C-01, a CODEOWNERS file, a PR template and branch rules                         | Checkstyle config, CODEOWNERS, PR template, branch rules     | C-01 check rewritten as a token rule after a false positive                    | Branch protection tested by a direct push, which was blocked and logged   |
| 5 | 2026-10-04 | Claude Opus 5 | Business Rules Register                              | Asked to turn each module's rules into numbered rules with boundary values for test design                                     | 43 numbered rules with boundary values                       | Numeric thresholds reviewed and accepted unchanged by the team                 | Boundary values cross-checked against the RTM test design                 |

|   |            |                 |                                                            |                                                                                                                           |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                             |                                                                                                                                                            |
|---|------------|-----------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | 2026-10-04 | Claude Opus 5   | Build and CI scaffold                                      | Asked for a Gradle build, a failing <code>MoneyTest</code> first, then <code>Money</code> , and a GitHub Actions workflow | Gradle build, <code>Money</code> , <code>MoneyTest</code> , GitHub Actions workflow                                                                                                                                                                                                                             | Written as a real TDD cycle, red commit before green                                                                                                                                                                                                                                                        | CI green: compile, 11 tests, Checkstyle, coverage                                                                                                          |
| 7 | 2026-10-04 | Claude Opus 5   | Submission PDFs                                            | Asked for a pandoc/LaTeX script producing both submission PDFs with a title page                                          | Build script and title page                                                                                                                                                                                                                                                                                     | Fonts changed to carry mathematical symbols                                                                                                                                                                                                                                                                 | Pages inspected; a fault dropping <code>≤</code> and <code>×</code> was found and fixed                                                                    |
| 8 | 2026-10-07 | Claude Opus 5.5 | Pre-submission review of both documents and the repository | Asked to check both M1 documents against the joint guidelines and rubrics, list inconsistencies, then apply the fixes     | Corrections to both documents, RTM, risk register, business rules, README and CODEOWNERS: M1 Team Lead, class views matched to the text and RTM, SOLID status labels, milestone references matched to the updated guidelines, an independent fact-check pass, PDFs rebuilt with an HTML/CSS pipeline (Paged.js) | M1 Team Lead confirmed by the team; an interim four-milestone revision was reverted to three milestones when the instructor issued the updated guidelines the same day; risk-grade assignment and active-loan counting defined at the team lead's request and confirmed by all three members (BR-03, BR-09) | Each change checked against the guidelines PDF and the live repository (ruleset, CI runs, <code>Money.java</code> ); PDFs rebuilt and every page inspected |

## Declaration for Milestone 1

Each member named below has read, understood and verified the material credited to them, can explain and defend the design decisions, quality requirements and analysis it contains, and reviewed and modified any AI-assisted content where needed.

- **Muhammad Ibrahim — 2023446** · `origination`, `delinquency`; SQAP outline, risk register
- **Hassan Khalid — 2023242** · `products`, `reporting`; quality requirements, RTM
- **Tughrul Hussain — 2023532** · `ledger`; standards awareness, Cost of Quality, repository setup

## Team rules for AI use

1. Generated code enters the repository only through the module owner, who reviews it and answers for it.
2. No generated test is accepted without reading what it asserts.
3. Business rules and numeric thresholds are the team's decisions, not defaults.
4. Entries are added when the work happens, not before a deadline.
5. Anything a member cannot explain is rewritten or withdrawn before submission.